

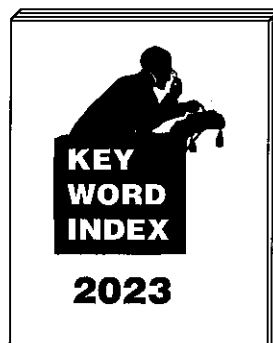
2023

JOURNEYMAN

OPEN

BOOK

EXAM #10



25 QUESTIONS

TIME LIMIT - 1 HOUR

TIME SPENT **MINUTES**

SCORE **%**

JOURNEYMAN OPEN BOOK EXAM #10 25 Questions One Hour Time Limit

1. Motors and other rotating electrical machinery shall be totally enclosed or designed so as to minimize the entrance of ____.

- (a) moisture (b) dust (c) corrosive particles (d) all of these

2. For the purpose of load calculations, the square footage of a dwelling unit includes ____.

- (a) open porches
(b) garages
(c) areas not adaptable as future occupiable space
(d) all of these

3. A receptacle's grounding terminal shall only be used for connection to the ____ conductor.

- (a) grounded (b) equipment grounding (c) ungrounded (d) grounding electrode

4. Where overcurrent protection is provided as part of the industrial control panel, the supply conductors shall be considered as either feeders or ____ as covered by 240.21.

- (a) taps (b) branch circuits (c) shunts (d) isolated

5. Backfill that contains large rocks, paving materials, cinders, large or sharply angular substances, or corrosive material shall not be placed in an excavation where materials may damage raceways, cables, conductors, or other structures or prevent adequate compaction of fill or contribute to the ____ of raceways, cables, or other substructures.

- (a) damage (b) deterioration (c) aging (d) corrosion

6. _____. An electric device and circuit that controls a disconnecting means through a relay or equivalent device.

- (a) Remote Disconnect Control (b) Safety Switch (c) Knief Switch (d) Shunt Trip

7. Which of the following statements is false?

- (a) For isolated ground receptacles, the receptacle grounding terminal is required to be connected to an insulated equipment grounding conductor run with the circuit conductors.
(b) Round boxes are required to be used where conduits or connectors requiring the use of locknuts or bushings are to be connected to the side of the box.
(c) Control circuit devices with screw-type pressure terminals used with #14 or smaller copper conductors are required to be torqued to a minimum of 7 lb-in.
(d) Type MC cable is permitted to be manufactured as a single-conductor cable.

8. A permanent sign shall be attached to the cable tray at intervals not greater than 20 feet. The sign shall read as follows:

- (a) **CABLE TRAY FOR WELDING CABLES ONLY**
- (b) **CABLE TRAY MAXIMUM VOLTAGE 300**
- (c) **THE CABLE TRAY PROVIDES SUPPORT AT 6" INTERVALS**
- (d) **CABLES SHALL BE LOW VOLTAGE**

9. _____. The individual responsible for starting, stopping, and controlling an amusement or supervising a concession.

- (a) **Ticket Taker**
- (b) **Assistant**
- (c) **Operator**
- (d) **Manager**

10. A commercial building has an opening door in the wall on the second floor through which materials are moved into and out of a storage area located on the second floor. An overhead 120v branch circuit is to be installed from the second floor area of the building to a pole on which a flood-light will be mounted. Which of the following best describes Code requirements for the installation of this branch-circuit?

- (a) **It is not permitted to be installed overhead.**
- (b) **It must be at least 3' from the side of the door.**
- (c) **It must be at least 3' above the bottom of the door.**
- (d) **It must not obstruct entrance to the material handling door and be 3' from the door.**

11. Outdoor self-supporting radio and television receiving antennas, such as vertical rods, dishes, etc., shall be located well away from overhead conductors of electric light and power circuits of over ____ volts to ground.

- (a) **30**
- (b) **120**
- (c) **150**
- (d) **300**

12. A Class 1 power-limited circuit shall be supplied from a source having a rated output of not more than 30 volts and ____ volt amperes.

- (a) **1000**
- (b) **1200**
- (c) **1500**
- (d) **2000**

13. Enclosures that are not over 100 cubic inches in size and which have two conduits supported within three feet on either side of the enclosure and the enclosure does not contain devices or support fixtures shall not be required to have the enclosure supported if the conduits are ____.

- (a) **rigid nonmetallic conduits**
- (b) **threaded into hubs identified for the purpose**
- (c) **installed with locknuts inside and outside enclosure**
- (d) **shoulders of fittings outside and locknuts inside the box**

14. Size #18 or #16 fixture wires, and flexible cords shall be permitted for the control and operating circuits of X-ray and auxiliary equipment where protected by not larger than ____ ampere overcurrent devices.

(a) 6 (b) 15 (c) 20 (d) 30

15. Conductors drawn from a copper-clad aluminum rod with the copper metallurgically bonded to an aluminum core. The copper forms a minimum of ____ percent of the csa.

(a) 10 (b) 20 (c) 40 (d) 70

16. Concrete, brick or tile walls are considered as being ____.

(a) isolated (b) insulators (c) grounded (d) dry locations

17. ____ locations are those that are hazardous because of the presence of easily ignitable fibers or flyings.

(a) Class I (b) Class II (c) Class III (d) Class II, division II

18. An isolated ground receptacle shall be identified by ____.

**(a) CO/ALR orange marking on the face of the receptacle
(b) a metal faceplate not less than 0.030 inches in thickness
(c) a reset type test button on the face of the receptacle
(d) an orange triangle located on the face of the receptacle**

19. In a hospital General Care Area, each patient bed location shall be provided with a minimum of ____ receptacles.

(a) 3 single (b) 2 single or 1 duplex (c) 2 duplex or 4 single (d) 4 duplex or 8 single

20. Balconies, decks, and porches that are attached to the dwelling unit and are accessible from inside the dwelling unit shall have at least one receptacle outlet accessible from the balcony, deck, or porch. The receptacle outlet shall not be located more than ____ above the balcony, deck, or porch walking surface.

(a) 6' (b) 6' 6" (c) 7' (d) 8'

21. Energized equipment parts for fire pump installations are required to be located at least ____ inches above floor level.

(a) 6" (b) 8" (c) 10" (d) 12"

22. Conductors having ____ insulation and operating at different voltage levels shall not occupy the same enclosure, cable, or raceway.

(a) thermoset (b) flame-retardant (c) nonshielded (d) silicone rubber

23. In a recreational vehicle park with electrical supply, at least ____ % of the sites shall be equipped with 30 ampere, 125 volt receptacles.

(a) 25 (b) 30 (c) 40 (d) 70

24. The conductor used to connect the grounded circuit of a wiring system to a grounding electrode is the ____.

**(a) grounded conductor (b) bonding jumper
(c) main bonding jumper (d) grounding conductor**

25. Backfill that contains large rocks, paving materials, cinders, large or sharply angular substances, or corrosive material shall not be placed in an excavation where materials can ____ raceways, cables, or other structures or prevent adequate compaction of fill or contribute to the corrosion of raceways, cables, or other substructures over 1000v.

(a) damage (b) deteriorate (c) age (d) corrode